

SamPave Design Auditor & Checker Validation Checklist

CRITICAL SAFETY DISCLAIMER

IMPORTANT:

SamPave is a professional decision-support engineering tool. All empirical design projections, thickness suggestions, material specifications, and mechanistic calculations generated by this software are for engineering screening and decision-support only. Final pavement designs, thickness profiles, and mix configurations must be independently verified, reviewed, and signed off by a licensed and qualified Professional Pavement Engineer before field execution, bidding, or construction.

This checklist is designed for senior design auditors, peer checkers, and government examiners to verify the integrity of pavement designs processed using the SamPave Engineering Suite.

1. Input Verification Check

- [] **Classified Traffic Volume Counts:** Verify that initial CVPD is sourced from a 7-day volume count survey (per IRC:9) and has been adjusted for the years between count and construction completion.
- [] **Traffic Growth Rate:** Confirm the annual growth rate ($r\%$) matches local masterplans or econometric projections. Watch out for unrealistically high projections ($r > 8.0\%$).
- [] **Design Life (n):** Verify design life matches road category specifications (typically 15 years for National Highways/State Highways, 20 years for expressways).
- [] **Vehicle Damage Factor (VDF):** Cross-check VDF against regional axle-load survey studies. If default tables are used, confirm the traffic volume class matches MORTH guidelines.
- [] **Lane Distribution Factor (LDF):** Confirm LDF matches road configuration (e.g. 0.75 for 4-lane divided carriageway).
- [] **Soaked CBR:** Confirm the subgrade soil CBR value represents a laboratory 4-day soaked test (per IS:2720 Part 16) on the worst-performing borrow-pit soil samples.

2. Structural Layer Compatibility Check

- [] **Base-to-Subbase Modular Ratio:** Confirm that the elastic modulus ratio between WMM (granular base) and GSB (granular sub-base) is between 2.0 and 4.0.
- [] **Stiffness Inversion:** Verify that lower layers are not stiffer than upper layers (e.g., granular sub-base modulus must not exceed granular base modulus).
- *Exception:* Bituminous cover over a Cement-Treated Base (CTB) slab is not considered a stiffness inversion.

- [] **Stiff CTB Over Granular Layer:** If a Cement-Treated Base ($E \geq 5000$ MPa) is specified, confirm that the underlying sub-base is either a Cement-Treated Sub-base (CTS) or a high-strength GSB ($E \geq 500$ MPa). If placed directly on weak support ($E < 500$ MPa), confirm shrinkage-relief measures are detailed.
 - [] **Poisson's Ratio:** Confirm Poisson's ratios conform to typical elastic limits:
 - Bituminous Concrete / DBM: 0.35
 - Wet Mix Macadam / Granular Base: 0.35
 - Cement-Treated Base / Sub-base: 0.25
 - Subgrade Soil: 0.35
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3. Mix Design Volumetrics & OBC Check

- [] **Air Voids (V_a):** Confirm air voids at the designated Optimum Binder Content (OBC) fall within the target design specifications (3.0% to 5.0%, typically 4.0% for DBM/BC).
 - [] **Void in Mineral Aggregate (VMA):** Verify VMA complies with minimum requirements (typically $\geq 12.0\%$ for DBM with 37.5mm nominal size and $\geq 13.0\%$ for 26.5mm size, per MORTH Table 500-11).
 - [] **Void Filled with Bitumen (VFB):** Check that VFB is within the specified range (65% to 75% for DBM/BC).
 - [] **Stability & Flow:** Verify stability is > 9.0 kN and flow is between 2.0 mm and 4.0 mm.
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4. IITPAVE Mechanistic Verification

- [] **Production Mode:** Confirm the design has been processed in **Mechanistic Verified Mode** (via local `IITPAVE.exe` runs) rather than Decision Support fallback mode.
 - [] **Tensile Strain (ϵ_t):** Confirm the tensile strain at the bottom of the bituminous layer does not exceed the fatigue life design strain.
 - [] **Vertical Compressive Strain (ϵ_v):** Confirm the vertical strain at the top of the subgrade does not exceed the rutting life design strain.
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5. Review Checklist and Project Sign-Off

- [] **Project Lock State:** Verify the project is **Locked** in the Submission Center to prevent further modifications.
- [] **Submission ZIP Package:** Verify the ZIP archive exists and contains all required output logs and calculations.
- [] **Independent Engineer Sign-off:** Ensure the final report bears the manual signature and seal of a licensed Professional Pavement Engineer.